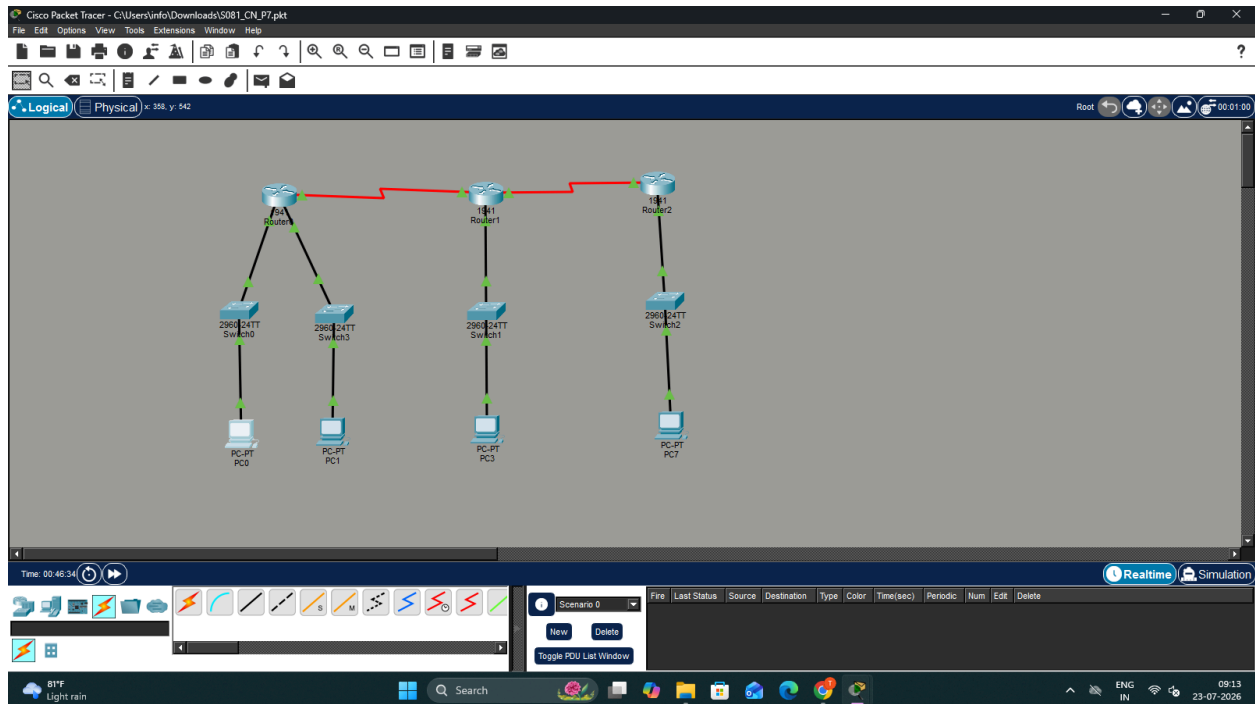


PRACTICAL NO: 4 M.V.L.U

Aim :- Refer the snap create the network in packet tracer and analyze the route using RIP Protocol.



Router1

```
Router0
Physical Config CLI Attributes
GigabitEthernet0/1 192.168.14.1 YES manual administratively down down
Serial0/1/0 192.168.10.1 YES manual up
Serial0/1/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/1
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#
%LINK-6-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router#
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.168.11.0 255.255.255.0 192.168.10.2
Router(config)#ip route 192.168.12.0 255.255.255.0 192.168.10.2
Router(config)#ip route 192.168.16.0 255.255.255.0 192.168.10.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Building configuration...
[OK]
Router#
Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, S - SGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
C 192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.10.0/24 is directly connected, Serial0/1/0
L 192.168.10.1/32 is directly connected, Serial0/1/0
S 192.168.11.0/24 [1/0] via 192.168.10.2
S 192.168.12.0/24 [1/0] via 192.168.10.2
C 192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.13.0/24 is directly connected, GigabitEthernet0/0
L 192.168.13.1/32 is directly connected, GigabitEthernet0/0
C 192.168.14.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.14.0/24 is directly connected, GigabitEthernet0/1
```

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The screenshot shows the configuration window for Router0, specifically the GigabitEthernet0 interface. The left sidebar contains a tree view with categories: GLOBAL, Settings, Algorithm Settings, ROUTING, Static, RIP, SWITCHING, VLAN Database, and INTERFACE. Under the INTERFACE category, GigabitEthernet0/0 is selected. The main area displays the configuration for GigabitEthernet0/0, including Port Status (On), Link Speed (1000 Mbps), Duplex (Full Duplex), MAC Address (000A.F34D.8501), and IP Configuration (IPv4 Address: 192.168.13.1, Subnet Mask: 255.255.255.0). The Tx Ring Limit is set to 10. Below the configuration area, a command window shows the following commands:

```
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/1/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/1/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
```

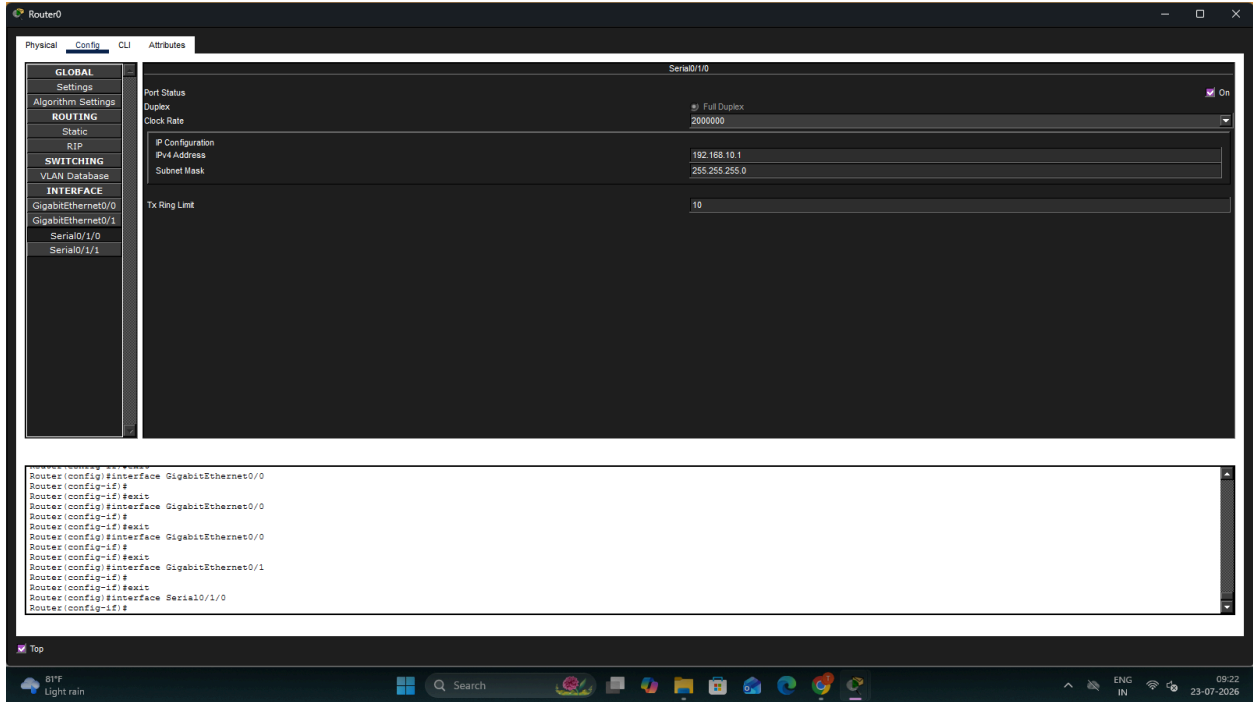
The bottom of the window shows a taskbar with a weather widget (81°F, Light rain), a search bar, and system icons (ENG, IN, 09:20, 23-07-2026).

The screenshot shows the configuration window for Router0, specifically the GigabitEthernet1 interface. The left sidebar is the same as in the first screenshot. Under the INTERFACE category, GigabitEthernet0/1 is selected. The main area displays the configuration for GigabitEthernet0/1, including Port Status (On), Link Speed (1000 Mbps), Duplex (Full Duplex), MAC Address (000A.F34D.8502), and IP Configuration (IPv4 Address: 192.168.14.1, Subnet Mask: 255.255.255.0). The Tx Ring Limit is set to 10. Below the configuration area, a command window shows the following commands:

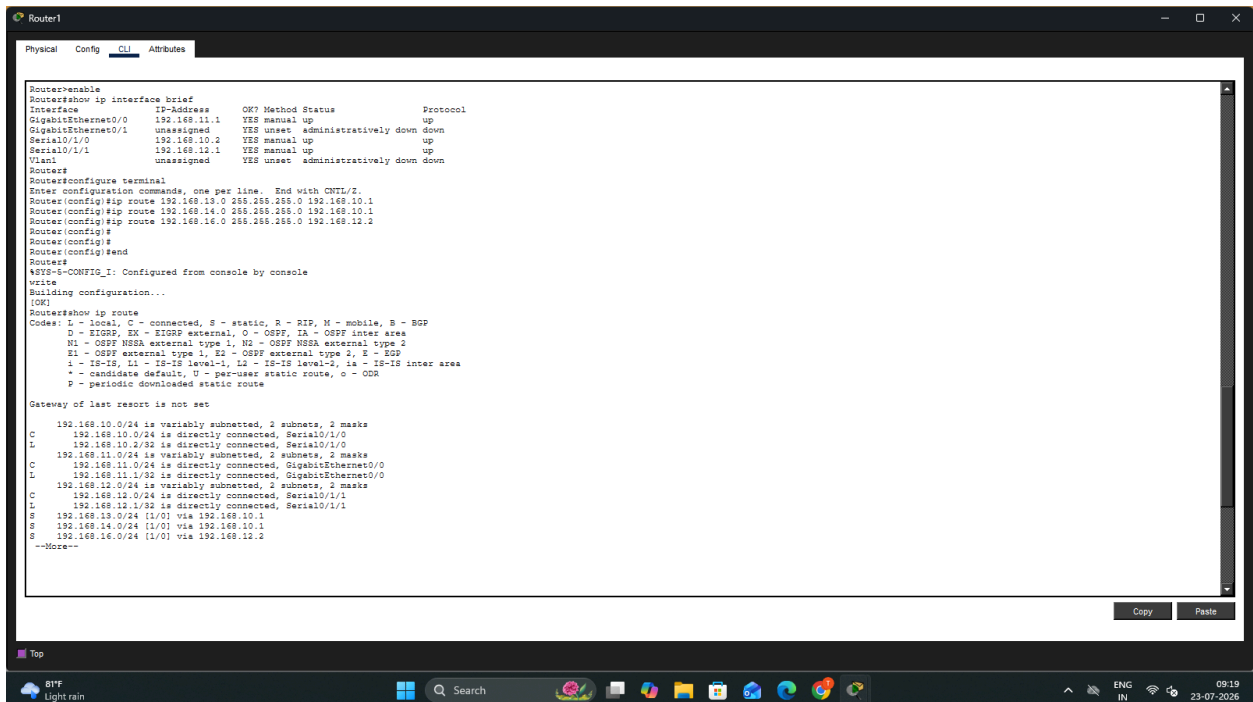
```
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
```

The bottom of the window shows a taskbar with a weather widget (81°F, Light rain), a search bar, and system icons (ENG, IN, 09:21, 23-07-2026).

PRACTICAL NO: 4 M.V.L.U



Router 2



PRACTICAL NO: 4 M.V.L.U

The screenshot shows the configuration page for Router1, specifically for the GigabitEthernet0/0 interface. The left sidebar contains a tree view with categories: GLOBAL, Settings, Algorithm Settings, ROUTING, Static, RIP, SWITCHING, VLAN Database, and INTERFACE. Under INTERFACE, the following interfaces are listed: GigabitEthernet0/0, GigabitEthernet0/1, Serial0/1/0, and Serial0/1/1. The main area displays the configuration for GigabitEthernet0/0, which is currently 'On'. The configuration includes: Port Status (On), Link Speed (1000 Mbps), Duplex (Full Duplex), MAC Address (00E0.BFA4.D401), P Configuration (192.168.11.1), Subnet Mask (255.255.255.0), and Tx Ring Limit (10). Below the configuration area is a command window showing the following commands:

```
Router>enable
Router#
Router>configure terminal
Router(config)#interface Serial0/1/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/1/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
```

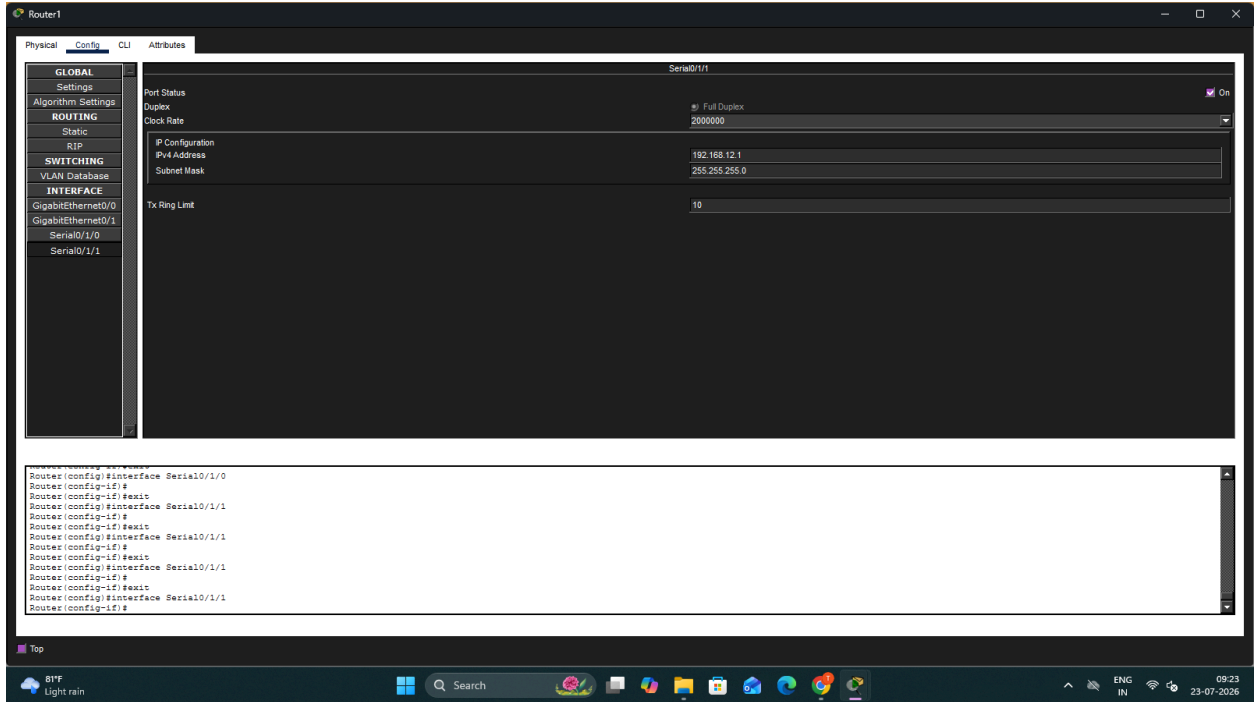
The bottom status bar shows the system clock as 09:22 on 23-07-2026, with weather information (81°F, Light rain) and system icons.

The screenshot shows the configuration page for Router1, specifically for the Serial0/1/0 interface. The left sidebar is identical to the previous screenshot. The main area displays the configuration for Serial0/1/0, which is currently 'On'. The configuration includes: Port Status (On), Duplex (Full Duplex), Clock Rate (2000000), P Configuration (192.168.10.2), Subnet Mask (255.255.255.0), and Tx Ring Limit (10). Below the configuration area is a command window showing the following commands:

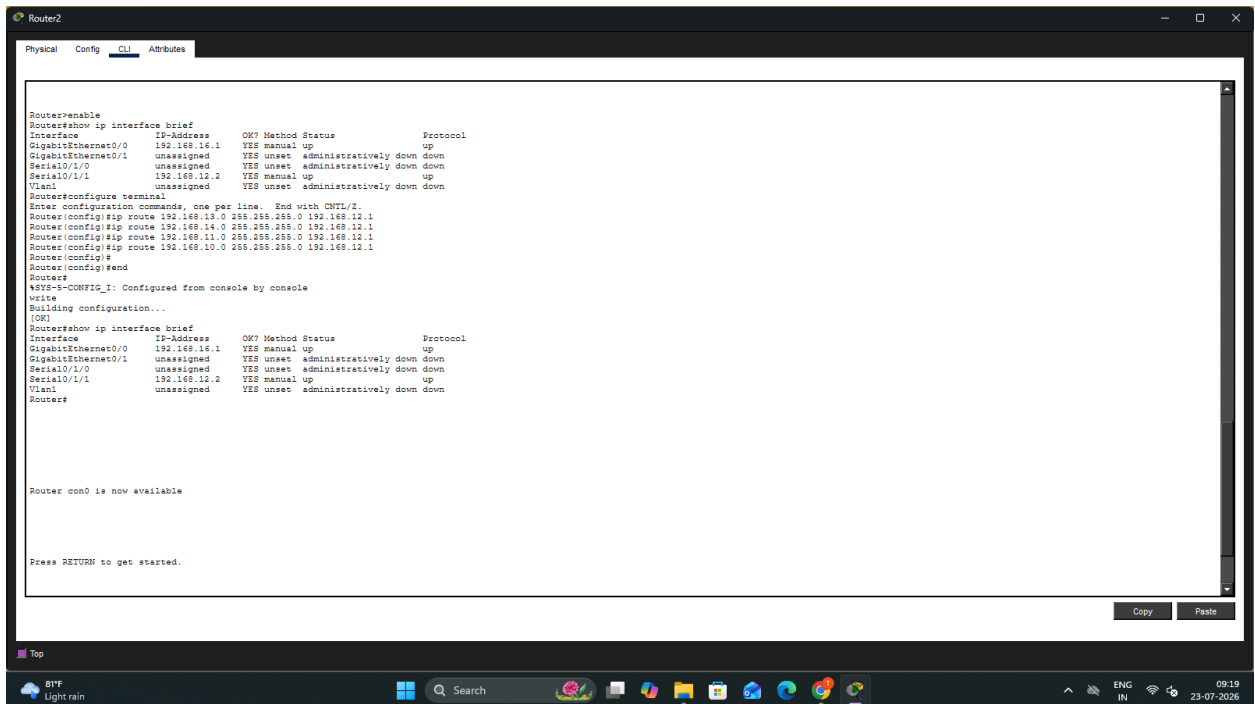
```
Router>configure terminal
Router(config)#interface Serial0/1/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/1/0
Router(config-if)#
```

The bottom status bar shows the system clock as 09:22 on 23-07-2026, with weather information (81°F, Light rain) and system icons.

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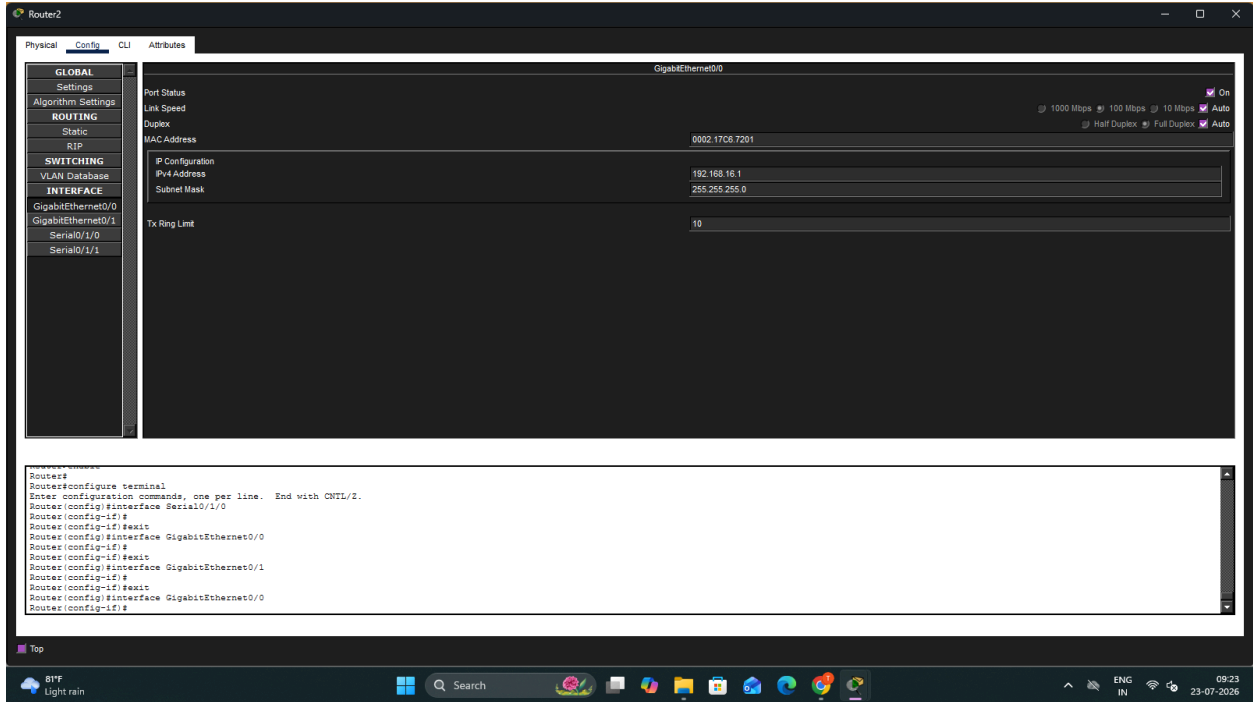


Router 3

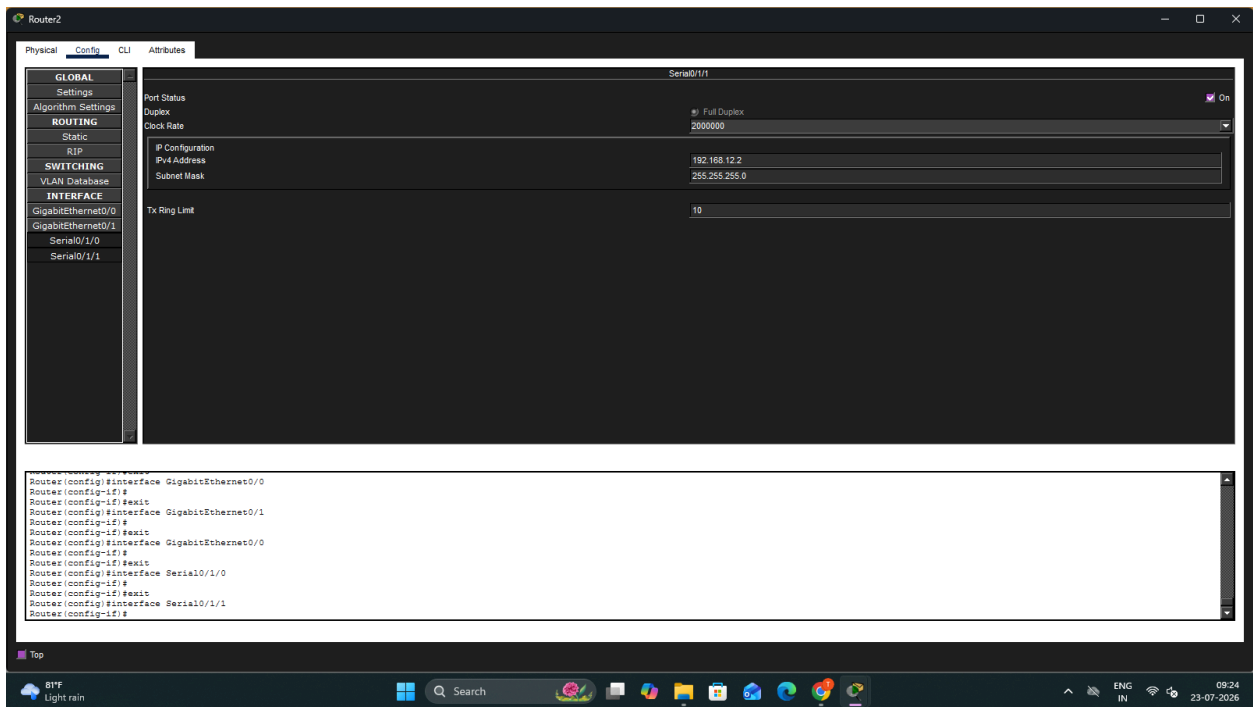


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PRACTICAL NO: 4 M.V.L.U

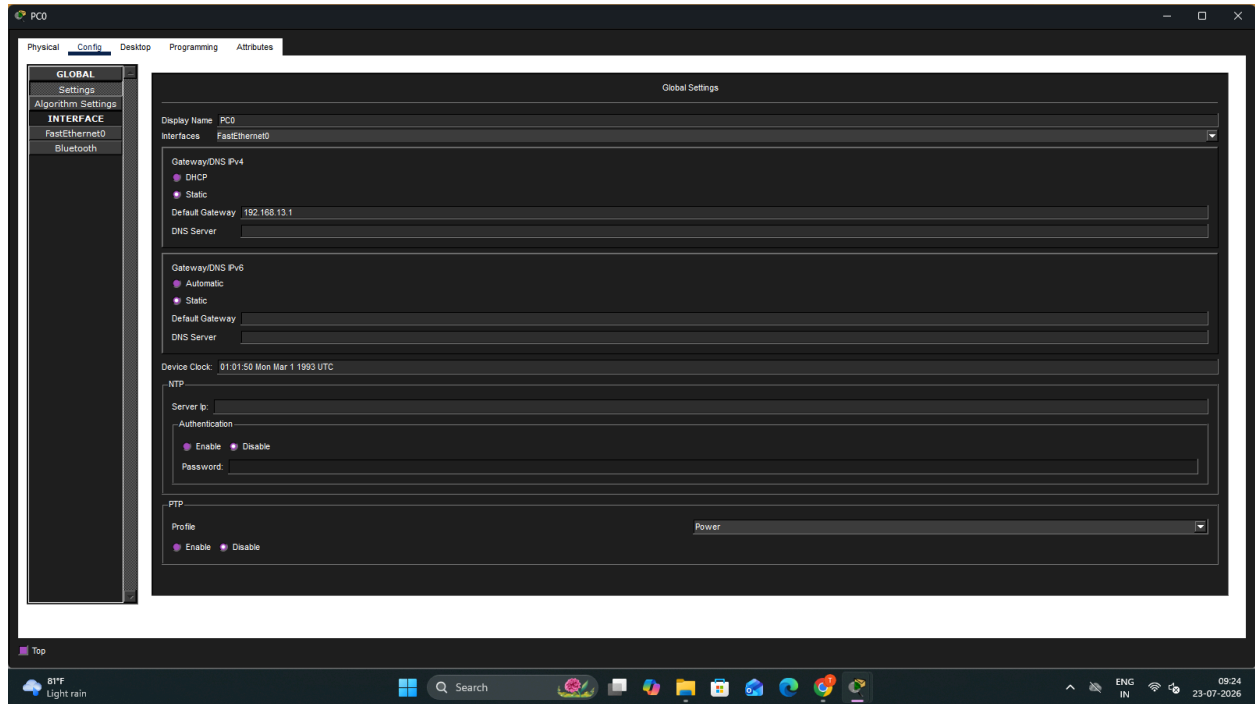


Pc 0

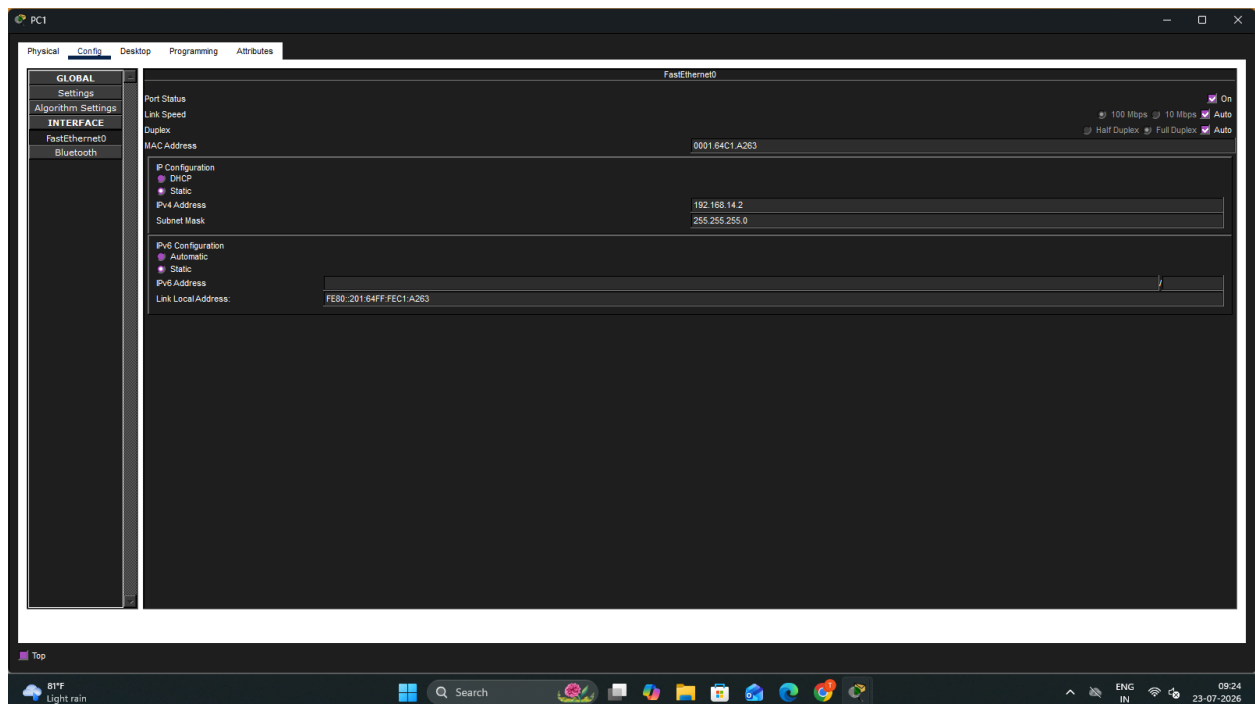


SANJANA SAHU T110

PRACTICAL NO: 4 M.V.L.U

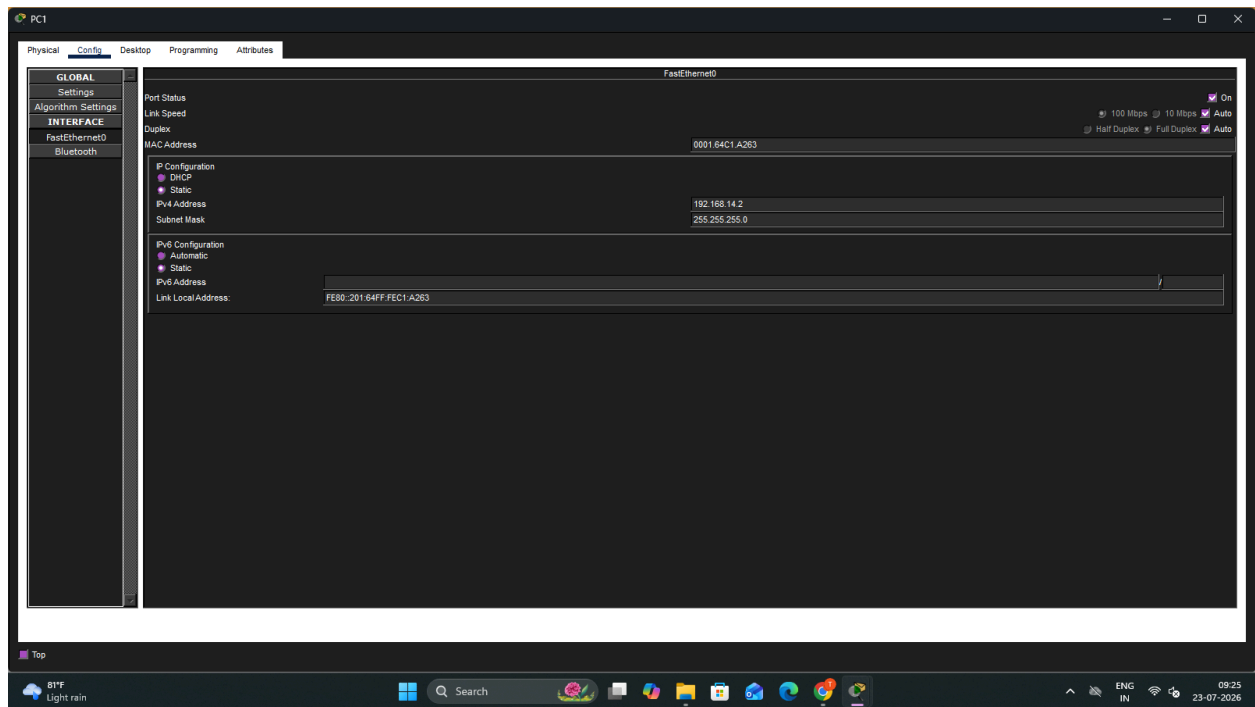
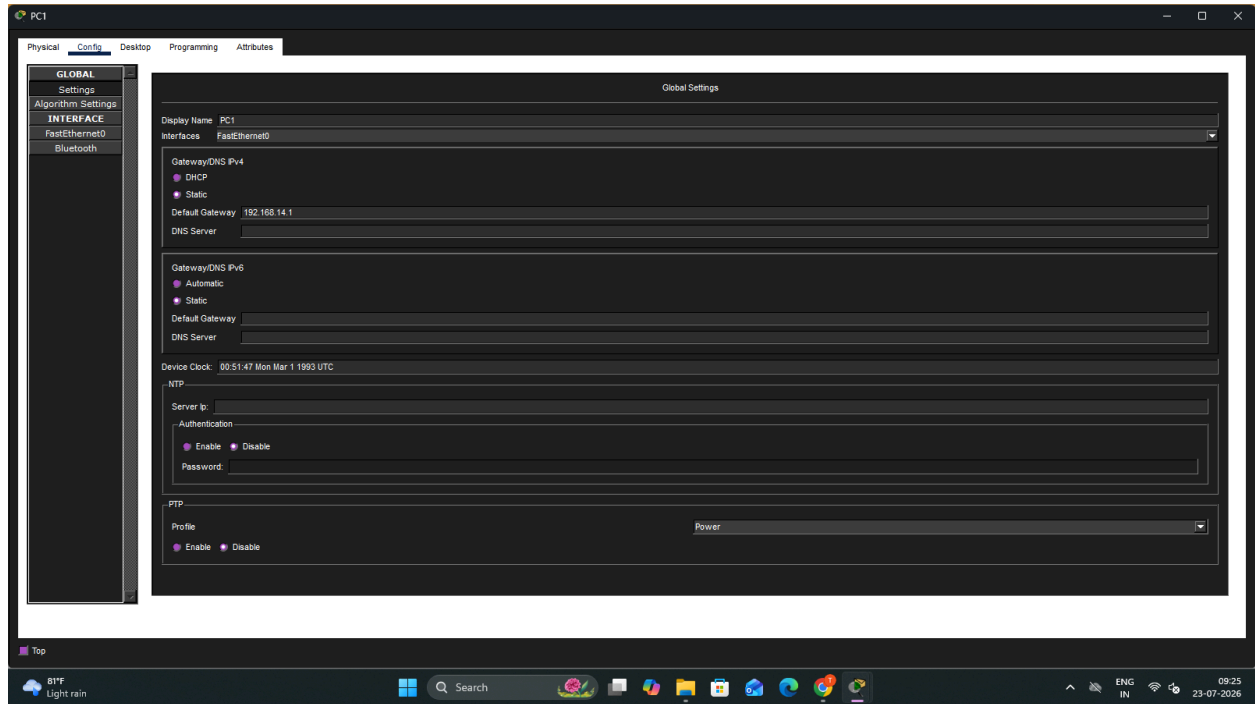


Pc 1



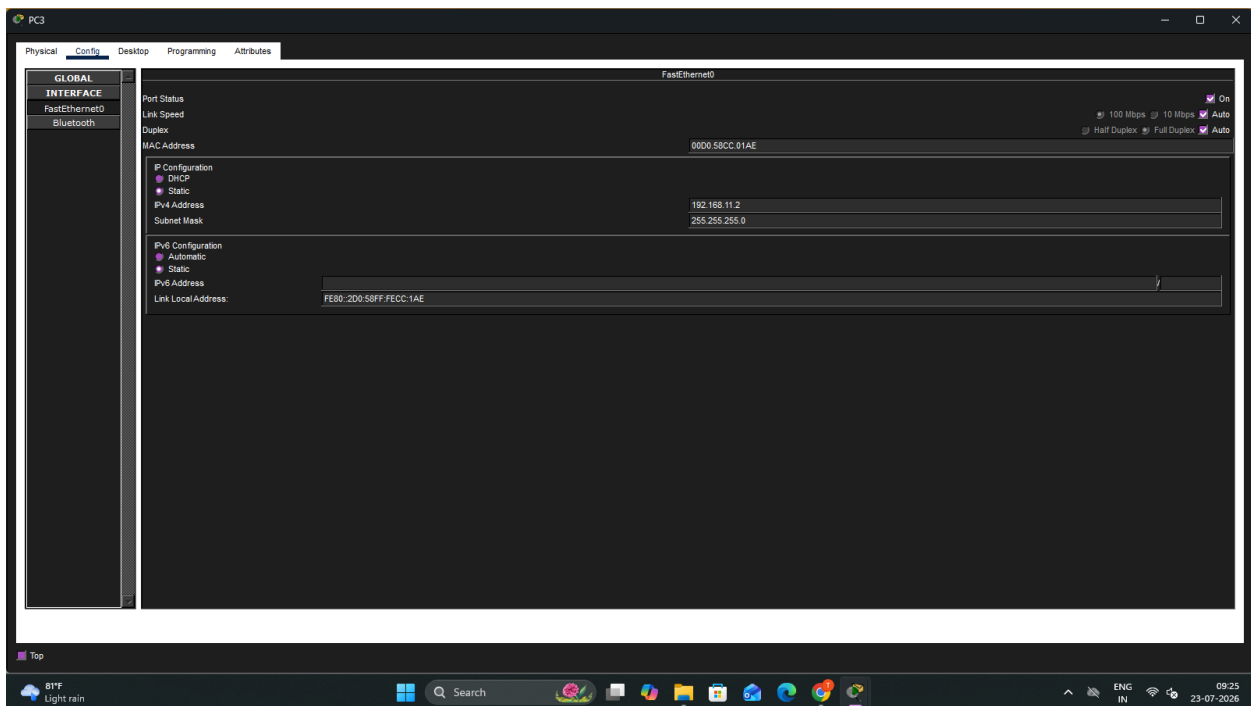
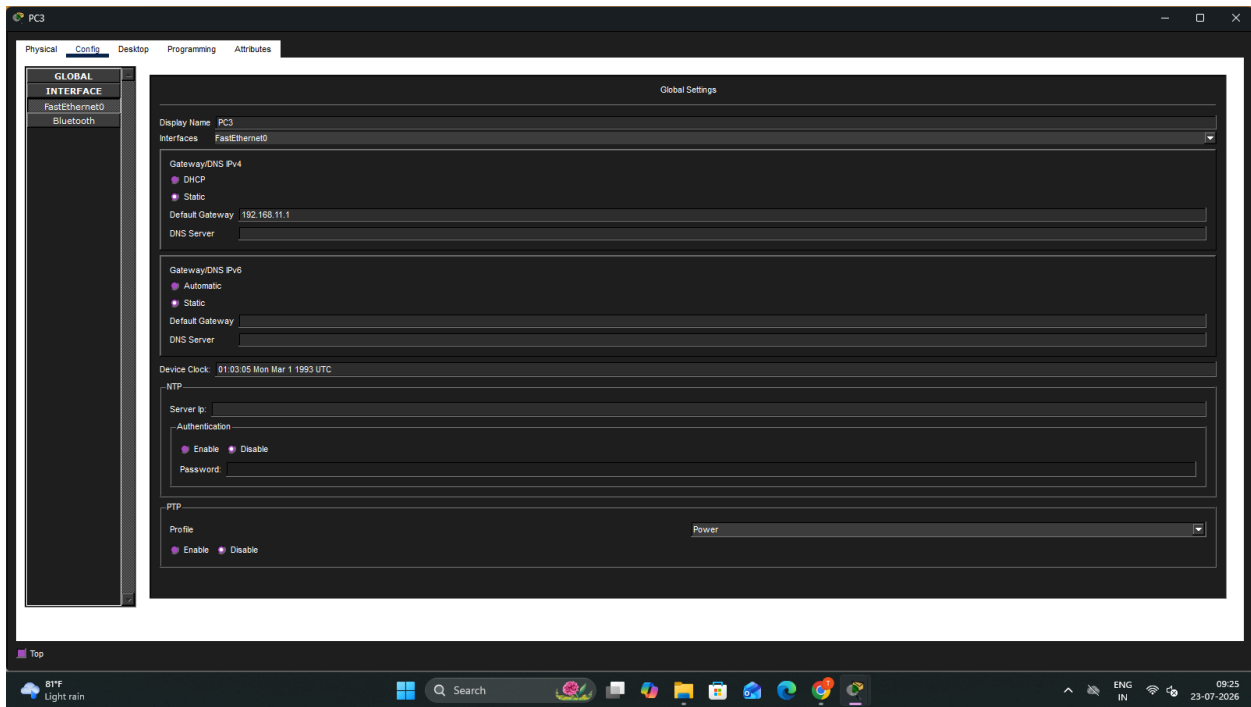
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PRACTICAL NO: 4 M.V.L.U

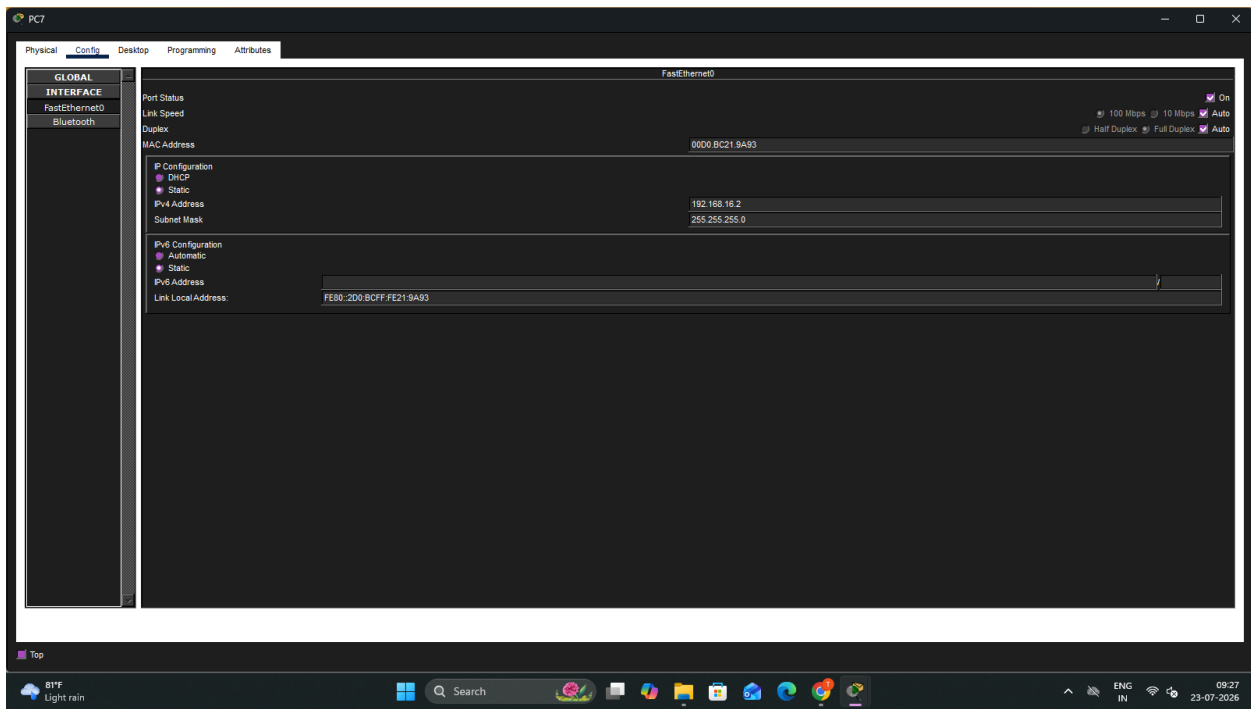
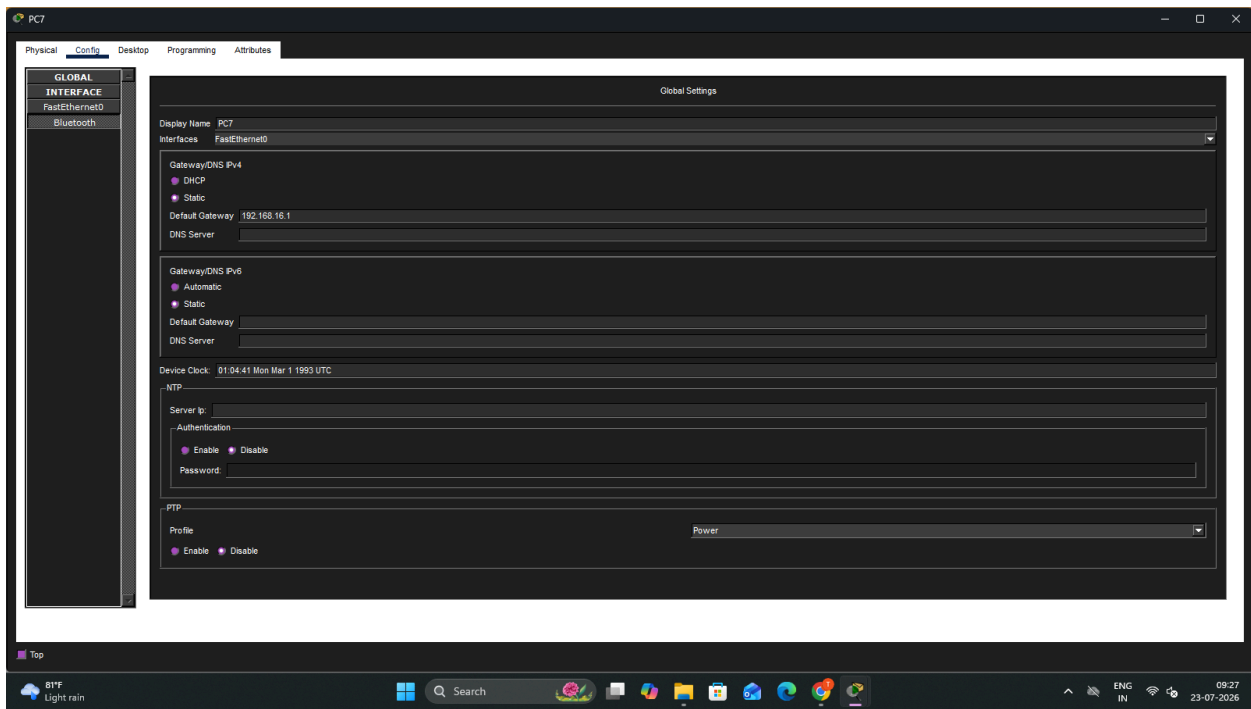
Pc 3



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PRACTICAL NO: 4 M.V.L.U

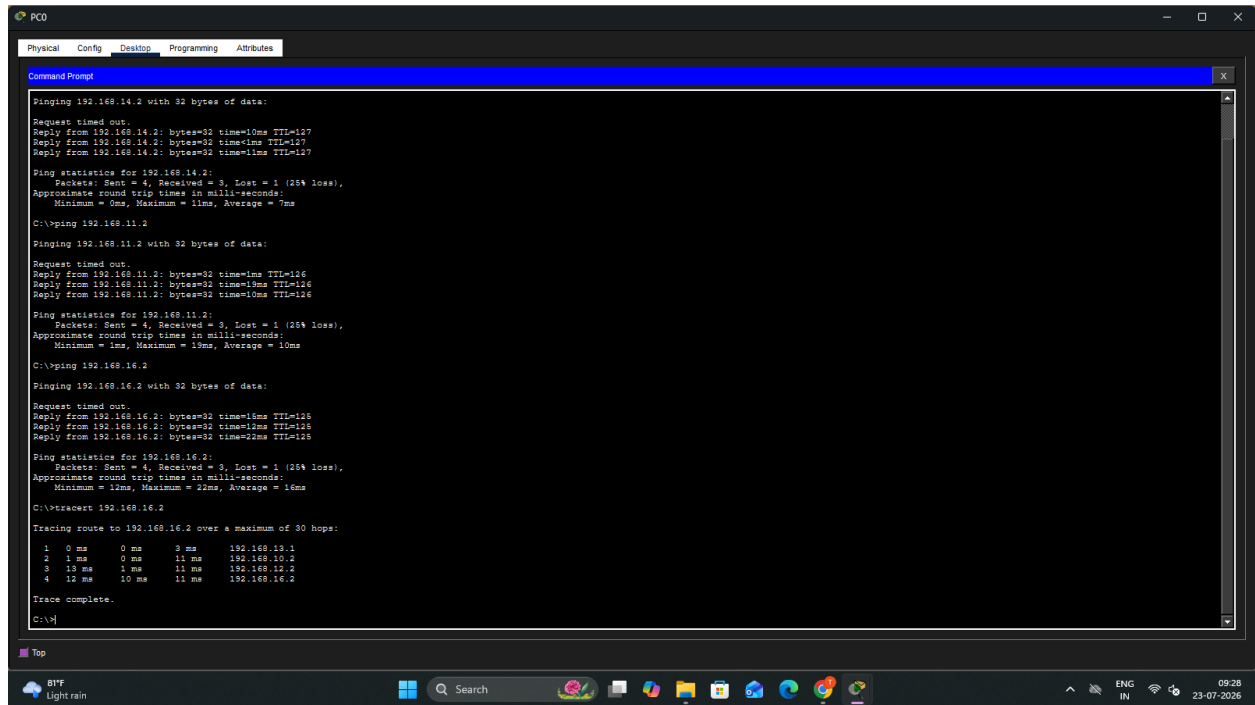
Pc 7



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PRACTICAL NO: 4 M.V.L.U

Ping response and traceroute



The screenshot shows a Windows PC0 interface with a Command Prompt window open. The Command Prompt displays the results of several ping and traceroute commands. The first ping is to 192.168.14.2, which times out. The second ping is to 192.168.11.2, which also times out. The third ping is to 192.168.16.2, which succeeds with a 100% success rate. The traceroute command is also run to 192.168.16.2, showing a successful path through three hops.

```
PC0
Physical  Config  Desktop  Programming  Attributes

Command Prompt

Pinging 192.168.14.2 with 32 bytes of data:
Request timed out.
Reply from 192.168.14.2: bytes=32 time=10ms TTL=127
Reply from 192.168.14.2: bytes=32 time<1ms TTL=127
Reply from 192.168.14.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.14.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 7ms

C:\>ping 192.168.11.2

Pinging 192.168.11.2 with 32 bytes of data:
Request timed out.
Reply from 192.168.11.2: bytes=32 time=1ms TTL=126
Reply from 192.168.11.2: bytes=32 time=13ms TTL=126
Reply from 192.168.11.2: bytes=32 time=10ms TTL=126

Ping statistics for 192.168.11.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 13ms, Average = 10ms

C:\>ping 192.168.16.2

Pinging 192.168.16.2 with 32 bytes of data:
Request timed out.
Reply from 192.168.16.2: bytes=32 time=15ms TTL=125
Reply from 192.168.16.2: bytes=32 time=12ms TTL=125
Reply from 192.168.16.2: bytes=32 time=22ms TTL=125

Ping statistics for 192.168.16.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 15ms, Maximum = 22ms, Average = 16ms

C:\>tracert 192.168.16.2

Tracing route to 192.168.16.2 over a maximum of 30 hops:
  0  0 ms    0 ms    3 ms    192.168.13.1
  1  1 ms    0 ms   11 ms    192.168.10.2
  2  13 ms   1 ms   11 ms    192.168.12.2
  3  12 ms   10 ms  11 ms    192.168.16.2
    Trace complete.

C:\>
```

Taskbar: BTf Light rain, Search, Task View, File Explorer, Microsoft Edge, Google Chrome, VLC media player, 09:28 23-07-2026